Workout 1B Homework 1 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:16

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

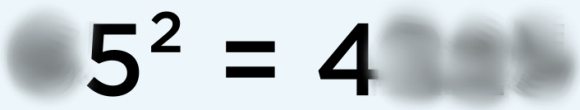
Review Later

Suppose $60^2 - 17^2 = 11X$. What is X?

2

Review Later

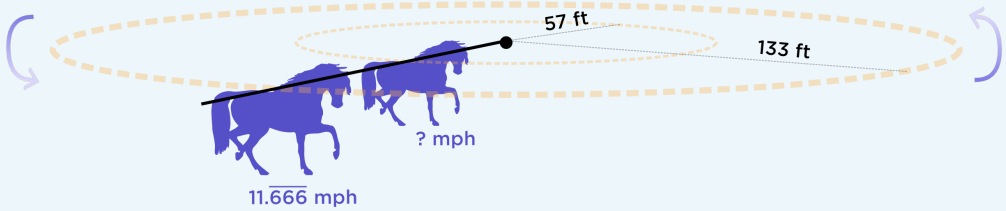
Prof. Loh spilled water on his Daily Challenge notes, so he can't see the first digit of the number that he was squaring. He only knows that it ended with a 5 and that its square was a four-digit number starting with 4. If the number he was squaring has the digits $x5$, what is the value of x?



3

Review Later

In the old days, horses used to drive mechanical machines like, for example, wheat threshers (hence the units of "horsepower" to measure engine power). Two horses are hooked up to a thresher, at distances of 57 feet and 133 feet from the center of the turnstile. If the outer horse is going at a speed of $11.\overline{666}$ miles per hour, how many miles per hour is the inner horse going?



4

Review Later

What is $\dfrac{2^{2^{2^2}}}{\left(\left(2^2\right)^2\right)^2}$?

5

Review Later

In terms of C, a temperature in degrees Celsius, what is the number of degrees of error in using the formula $F = 2C + 30$ to convert from celsius to fahrenheit, instead of the accurate formula $F = \frac{9}{5}C + 32$?

- $2C + 2$
- $\frac{C}{5}$
- $\frac{C}{2} - 5$
- $\frac{1}{5}C - 2$

6

Review Later

Alice and Bob are running on a track. Alice, who runs $6.\overline{666}\ldots\%$ faster than Bob, runs in the outer lane, which has a radius of 76 meters. Bob runs in the inner lane, which has a radius of 57 meters. In the time it takes Bob to run 5 laps, how many laps can Alice run?

7

Review Later

For what temperature in degrees Celsius does the estimation formula for converting celsius to fahrenheit, $F = 2C + 30$, give the correct temperature? Recall that the accurate formula for a temperature C in celsius and the same temperature F in fahrenheit is $F = \frac{9}{5}C + 32$.

8

Review Later

In Wonderland, a stolen cup case was under investigation. At the trial, the Hare stated that the Hatter had stolen the cup. Alice and the Hatter also gave testimony, but no one remembered what they said, and the record also disappeared. During the trial, it turned out that only one of the defendants had stolen the cup and that only that person had given truthful testimony. So who stole the cup?

- The Hare
- Alice
- The Hatter

9

Review Later

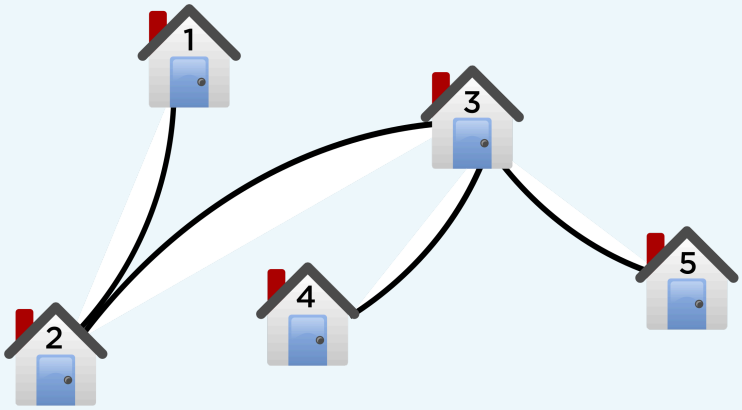
Given the facts below relating to the distances between pairs of houses, what is the distance between House 3 and House 5?

Distance between House 1 and House 4 = 75 meters

Distance between House 1 and House 5 = 73 meters

Distance between House 4 and House 5 = 34 meters

Distance between Houses 2 and 3 is 7 longer than the distance between Houses 1 and 2



10

Review Later

Recall that the reproduction rate of a disease is the average number of people that each contagious person infects. The reproduction rate for measles, the most infectious disease in history, was at least 12. Suppose that we start with one measles-infected person on Day 1, and assume that after one day, the contagious person is no longer contagious. Also assume that different people don't infect the same person. How many people will have ever been infected by Day 4?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying
them for the benefit of the challenge!

11

Review Later

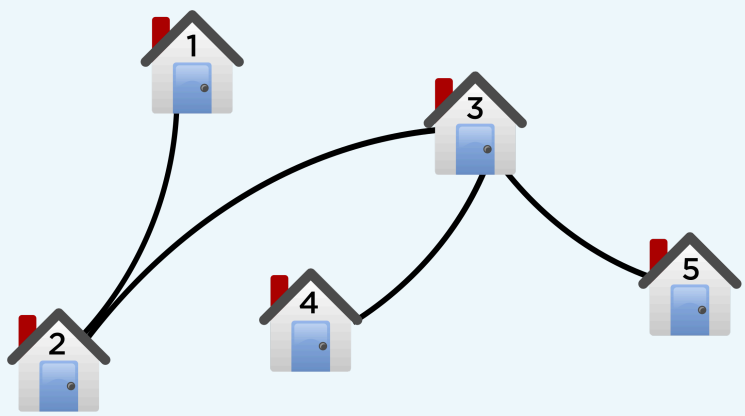
Given the facts below relating to the distances between pairs of houses, what is the distance between House 1 and House 2?

Distance between House 1 and House 4 = 75 meters

Distance between House 1 and House 5 = 73 meters

Distance between House 4 and House 5 = 34 meters

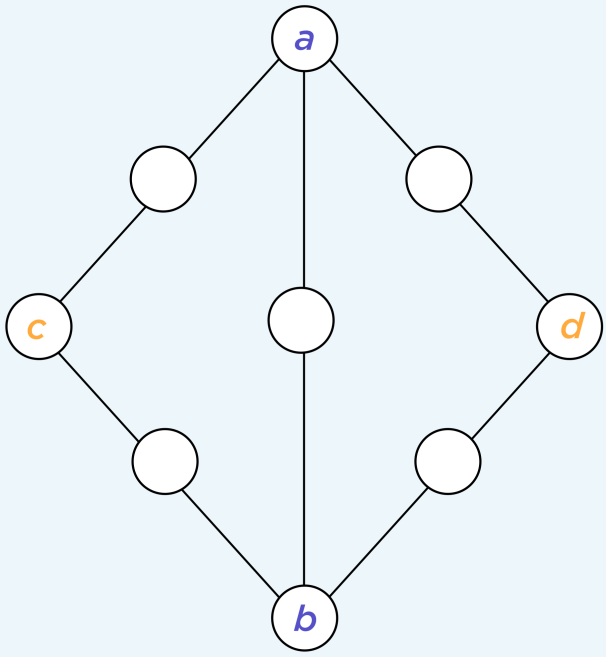
Distance between Houses 2 and 3 is 7 longer than the distance between Houses 1 and 2



12

Review Later

In the figure below, the numbers from 1 to 9 are used to fill in the circles, with each number occurring exactly once. Every line connecting three numbers has the same sum. If $a + b = 3$, what is the value of $c + d$?



13

Review Later

Suppose a virus has a reproduction rate ($R0$) equal to r. (Recall that this means that on average, each infected person infects r other people.) Assume that one person is infected on Day 1 and a person is only contagious for one day. In terms of r, how many people will be newly infected on Day n?

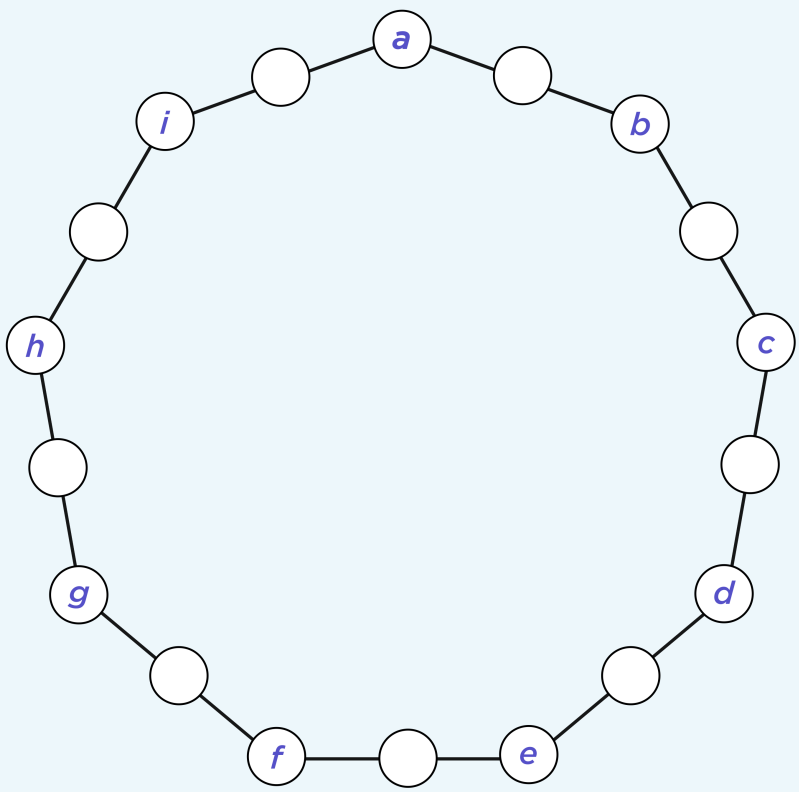
- $2r$
- n^r

- r^{n-1}
- nr
- $n + r$
- $1 + r + r^2 + \ldots + r^n$
- r^n

14

Review Later

In this nonagon, the integers from 1 to 18 are used to fill in the circles, with each number appearing once. The vertex numbers, $a, b, c, d, e, f, g, h,$ and i, are the numbers from 1 to 9. The three numbers on each side of the nonagon add up to the same sum, S. What is the value of S?



15

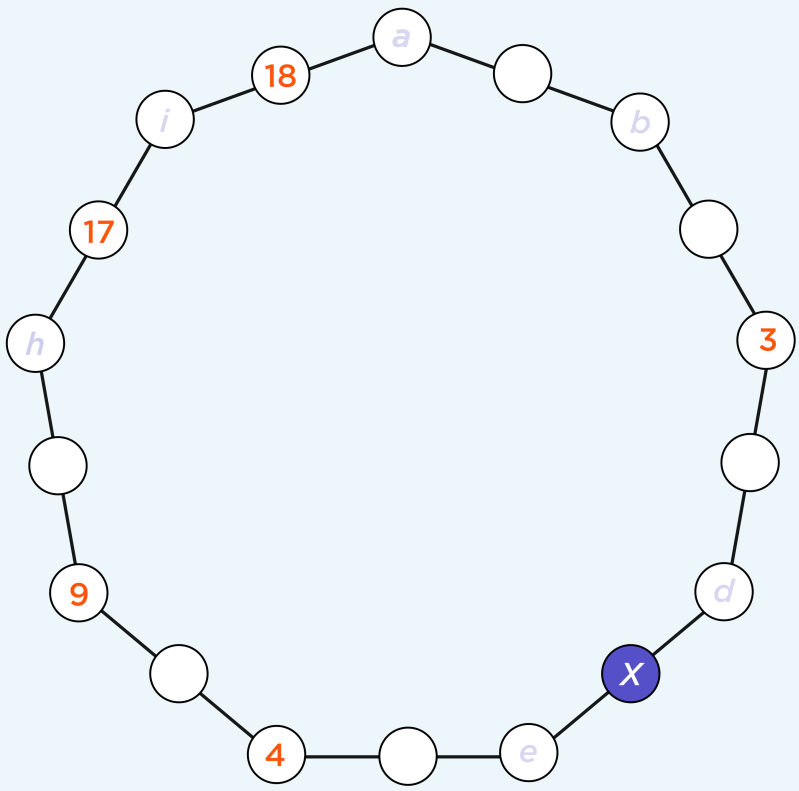
Review Later

Suppose that the reproduction rate ($R0$) for a disease equals 2, but suppose that people are contagious for 4 days instead of 1 day. Assume that no two people infect the same person. Now, after 4 days, how many people will have ever been infected?

16

Review Later

As in the previous question, the numbers from 1 to 18 are used to fill in the circles, with each number occurring exactly once. Every line connecting three numbers has the same sum. What is the value of the number in the bubble marked with an X?



17

Review Later

A disease starts spreading, with one person infected on the first day. Assume that people are contagious for only 1 day, and after 5 days, there were $1,555$ people who had ever been

infected. Based on the reproduction numbers given below, which disease was it?

- Smallpox $(R0 = 3.5)$
- Common Cold $(R0 = 1.8)$
- Chickenpox $(R0 = 10)$
- Rubella $(R0 = 7)$
- Polio $(R0 = 6)$
- Measles $(R0 = 12)$

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

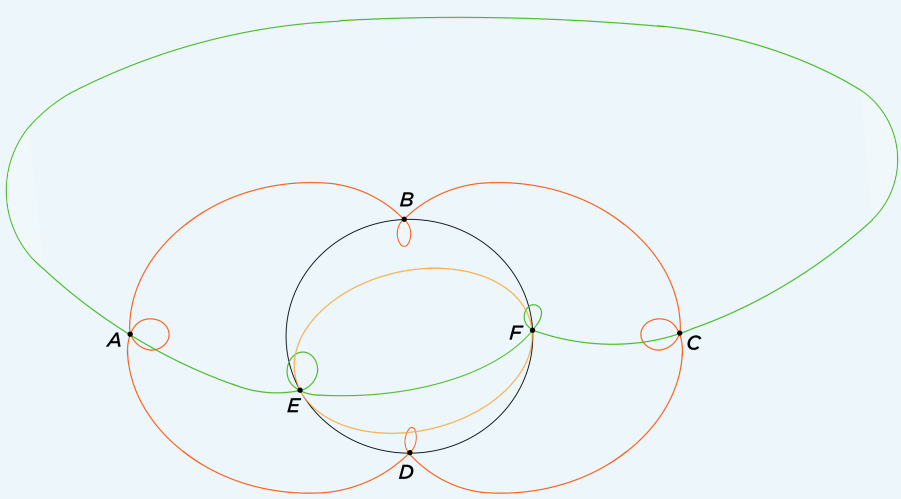
Alice and Bob are playing a game with a basket of 50 oranges. Starting with Alice, each player can take either $1, 2,$ or 3 oranges out of the basket. The last person to take an orange out of the basket loses, and the other player wins. Which player has a strategy that ensures they will always win the game?

- Alice
- Bob
- Neither

19

Review Later

In the squiggly paths of a game below, every time you pass a labeled mine (A, B, C, D, E or F), you get some gold, the amount specific to the particular mine. One full circuit of the red path gives 180 gold, one full circuit of the green path gives 92 gold, and one full circuit of the yellow path gives 22 gold. How much gold does one circuit around the black circle path give?



20

Review Later

Alice and Bob are playing a game with a basket of x oranges. Starting with Alice, each player can take either $1, 2,$ or 3 oranges out of the basket.
The last person to take an orange out of the basket wins. For how many values of x in $[100]$ (the integers from 1 to 100 inclusive) does Alice have a winning strategy?

View Marked Questions

Finished

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